

SYS::ONLINE · CONVERGENCE WINDOW OPEN · PLANETARY REGEN PROTOCOL ACTIVE

URTE

UNIVERSAL REGENERATION THERAPY ECOSYSTEM

"Regenerating life across scales — from cells to planets"

PHASE 1 2026–2029 · TRL PULL ACCELERATION · DUAL-USE: CLINICAL + PLANETARY · MOONSHOT TO 2040+

WHY NOW? THREE SIMULTANEOUS INFLECTIONS

PLANETARY BOUNDARIES

6 of 9 boundaries transgressed

- Biodiversity collapse accelerating
- Climate adaptation urgency critical
- Ecosystem services in multi-trillion \$ decline

Source: Richardson et al. 2023, IPBES, IPCC

HEALTHCARE CRISIS

\$10T+ annual global burden
chronic & degenerative disease

- Aging populations worldwide
- Late-stage intervention costs exploding
- Equity gaps in advanced therapies

Source: WHO 2024, ARM 2025

TECH CONVERGENCE

All domains at inflection simultaneously

- CRISPR/iPSC therapies approved (TRL 8-9)
- AlphaFold-class generative AI mature
- 3D bioprinting entering clinic
- ISRU (MOXIE) TRL 4-5 on Mars
- Digital twin regulatory pathways open

WHY INTEGRATION IS EXISTENTIAL — FOUR CRITICAL GAPS

REGENERATIVE MEDICINE

High TRL therapies exist (Casgevy etc.) but \$1–3M per treatment. Manufacturing bottlenecks & global equity crisis severe. No cross-learning with ecological systems.

ECOSYSTEM RESTORATION

Proven at project scale (reefs, mangroves, rewilding) but fragmented data standards, no standardized adaptive control, and zero shared simulation with clinical domains.

SYNTHETIC BIOLOGY & ISRU

Circuit designers and planetary modelers operate in complete isolation despite governing identical principles of feedback, resilience, and tipping points.

SPACE LIFE SUPPORT

NASA/ESA ISRU teams and clinical biomanufacturing both require closed-loop systems — yet share almost zero common tooling, validation frameworks or simulation environments.

URTE VISION — THE OPERATING SYSTEM FOR LIFE REGENERATION

A world — and eventually worlds — where damaged biological and ecological systems can be systematically regenerated using universal, science-based protocols, supported by intelligent orchestration that respects complexity, uncertainty, and ethical boundaries.

URTE is the critical integration layer — an 'ecosystem of ecosystems' that transforms siloed progress into coherent, multi-scale regenerative capability with a direct bridge to controlled planetary engineering.

01 UNIVERSAL ABSTRACTION

Scale-invariant mathematical & computational representations of living systems (molecular → planetary).

02 CLOSED-LOOP LEARNING

Every intervention generates data that immediately improves future decisions across medicine and ecology.

03 ETHICAL BY DESIGN

Multi-stakeholder oversight, reversibility, and planetary-boundary guardrails are non-negotiable architectural constraints.

04 OPEN STANDARDS + SAFEGUARDS

Data models, APIs & simulation kernels openly developed while protecting biosafety and strategic IP.

URTE HIGH-LEVEL LAYERED ARCHITECTURE — PRECISE CAPABILITY STACK

COMMAND & GOVERNANCE

Mission Control | Multi-Stakeholder Ethics Board | Regulatory Interface | Public Transparency Dashboards | International Treaty Interface

AI ORCHESTRATION & MULTI-SCALE SIMULATION

Generative Therapy/Protocol Designer | Predictive Digital Twins (Cell ↔ Planet) | Reinforcement Learning Optimizer | Physics-Informed Surrogates | Swarm Coordination Interface

BIOLOGICAL REGEN SUBSYSTEM

Cell/Gene Therapy Modules (TRL 6-9) | Tissue & Organ Engineering (TRL 4-7) | Personalized Medicine Pipelines | Advanced Bioprinting & Organoid Maturation

ECO-PLANETARY REGEN SUBSYSTEM

Habitat & Biodiversity Synthesis Engine | Soil/Air/Water Regeneration Protocols | Closed-Loop Life Support & ISRU Bio-Hybrid Simulation | Landscape-Scale Adaptive Management

SENSING, ACTUATION & DEPLOYMENT

Bioreactors, Bioprinters, Robotic Surgical Systems | IoT Sensor Swarms, Field Drones/Rovers, ISRU Processors | ****Bionanorobot Swarms**** (sandboxed precision repair at subcellular scale)

CONTINUOUS FEEDBACK & LEARNING LOOP

Real-time Telemetry → Model Update → Policy/Intervention Refine | Outcome Databases → Cross-Domain Learning → TRL Acceleration | Virtuous Evidence Flywheel

HOW URTE ACCELERATES REGENERATION — CORE ENABLERS & TRL PULL MECHANISM

CORE ENABLING TECHNOLOGIES

- Physics-Informed Neural Networks for real-time multi-scale digital twins (Raissi et al. 2019)
- Generative Design Engine for personalized therapies & eco-restoration plans
- Stochastic Optimal Control with embedded planetary-boundary & ethical guardrails
- Resilience Metric $R = -\lambda_{\text{dom}}$ (recovery speed after perturbation)
- Early-Warning Indicators: rising variance, lag-1 autocorrelation, skewness/kurtosis for tipping points
- Bionanorobot Swarms: targeted intracellular/tissue/ecosystem repair with kill-switches, real-time twin monitoring & progressive sandboxing (TRL 2-4 → 5-6)

TRL PULL STRATEGY — THE ACCELERATION FLYWHEEL

High-maturity clinical modules (approved cell/gene therapies TRL 8-9) generate immediate revenue, regulatory precedent, and rich longitudinal outcome data.

This subsidizes and de-risks lower-TRL modules through shared infrastructure:

- Mid-TRL bioprinting & organoids accelerated 4-6 → 6-7
- Ecosystem restoration protocols 5-7 → 7+
- Synthetic biology circuits & ISRU bio-hybrids 3-5 → 5-6
- Planetary-scale concepts via shared simulation kernels & governance scaffolding

Cross-domain learning is explicit: Clinical safety/dosing/immune data directly informs ecological models. Ecosystem resilience metrics improve clinical closed-loop systems.

Result: Compounding acceleration instead of isolated valleys of death.

THREE-PHASE STRATEGIC ROADMAP — PRECISE MILESTONES

PHASE 1 2026–2029

FOUNDATION & EARLY INTEGRATION

- Core modules advanced to TRL 6-7
- First AI-orchestrated human therapy pilots
- Terrestrial ecosystem restoration demonstrators
- 3-5 validated peer-reviewed use cases
- Open data standards + governance charter v1.0
- First cross-domain simulation validation published

KPI: ≥25% reduction in therapy development cycle time • Measurable improvement in restoration success rates • Shared ontology & simulation kernel v1.0 released

PHASE 2 2030–2034

CROSS-SCALE INTEROPERABILITY & ANALOG TESTING

- Full clinical ↔ ecological module interoperability
- Operational closed-loop digital twins (human + landscape)
- Rigorous Mars/Moon analog campaigns (Antarctic, Atacama, Arctic)
- Controlled bioregenerative life-support testbeds
- International ethics advisory board established

KPI: Documented cross-domain learning cases • ≥2 successful analog campaigns with quantifiable metrics • Regulatory acceptance of digital-twin evidence in ≥1 jurisdiction

PHASE 3 2035–2040+

FRONTIER DEMONSTRATION & GOVERNANCE MATURATION

- Controlled regional ecosystem engineering under international oversight
- Planetary regeneration concepts to TRL 4-5
- Orbital/lunar precursor bio-regen + ISRU-hybrid payloads flown
- Draft international protocol for planetary intervention endorsed by major spacefaring nations & relevant UN bodies

KPI: Measurable regional ecosystem indicator improvement in testbeds • Successful bio-regen + ISRU integration in space-relevant environments • Formal adoption/endorsement by ≥1 major international body

DUAL-USE MARKET OPPORTUNITY — CLINICAL + PLANETARY + SPACE

HUMAN HEALTH REGENERATION | Multi-hundred billion TAM

Cell/gene therapies, tissue engineering, personalized medicine pipelines. URTE reduces development cycle time $\geq 25\%$, lowers costs via high-fidelity digital-twin evidence, and improves global equity through modular open standards.

EARTH RESTORATION & ECOSYSTEM SERVICES | Multi-trillion annual ecosystem services market

Landscape-scale adaptive management, biodiversity credits, carbon markets, UN Decade on Ecosystem Restoration alignment. Standardized monitoring + closed-loop control creates entirely new asset classes and verification infrastructure.

SPACE & ISRU BIO-HYBRIDS | Artemis, Mars, commercial LEO & deep space programs

Life-support regeneration, bio-ISRU, closed-loop systems for long-duration missions. NASA/ESA contracts + commercial space operators. Direct precursor capability for controlled planetary engineering and eventual terraforming.

RESEARCHERS & INSTITUTIONAL GROUNDING — MOONSHOT CREDIBILITY

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"Regenerating life across scales — from cells to planets"

- ▶ Grounded rigorously in current-world Technology Readiness Levels (TRL 1–9) as of mid-2026 — no hype, pure engineering reality.
- ▶ Biomathematical foundations: dynamical systems theory, stochastic optimal control, tipping-point early-warning indicators, multi-scale coupled PDE digital twins.
- ▶ Architecture emphasizes ethical governance, multi-stakeholder oversight, reversibility-by-design, and planetary-boundary guardrails from first principles.
- ▶ Aligned with: Alliance for Regenerative Medicine, ISSCR Guidelines 2025, NASA Artemis & ISRU roadmaps, UN Decade on Ecosystem Restoration, Outer Space Treaty principles.
- ▶ Open platform model designed for contribution and adoption by academia, biotech, conservation organizations, space agencies (NASA, ESA, CNSA), regulators, and civil society.

THE ASK — FUND PHASE 1 TO UNLOCK THE MOONSHOT

URTE is the open, evolving platform ecosystem that will define how humanity regenerates life at every scale in the 21st century — and prepares the technical and ethical foundations for multi-planetary civilization.

We are seeking strategic partners and investors to fund Phase 1 execution:

- Build and deploy URTE core platform v1.0 (AI orchestration backbone, multi-scale digital twin engine, first bionanorobot integration sandbox)
- Execute validated pilots in personalized regenerative medicine and terrestrial ecosystem restoration with measurable outcomes
- Establish the international ethics & governance sandbox framework as global reference
- Accelerate TRL advancement across domains through shared infrastructure and deliberate cross-subsidization (TRL Pull)

The Convergence Window is open now. The foundational technologies exist. The ethical frameworks are being architected from day one.

Now we build the universal regeneration layer that turns planetary crisis into regenerative capability — for Earth and for the multi-planetary future we choose to create.

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CORE VALUE PROPOSITION — WHY URTE IS THE ONLY INTEGRATOR

SCALE-INVARIANT REGENERATION PROTOCOLS

Same mathematical and computational language for directed repair from intracellular networks to watersheds to regional biomes to planetary surfaces. No other platform offers this abstraction.

HIGH-FIDELITY MULTI-SCALE DIGITAL TWINS

Physics-informed neural networks enabling real-time predictive simulation and intervention optimization from cell to planet — with continuous model updating from real-world telemetry.

TRL PULL ACCELERATION ENGINE

High-maturity clinical modules generate revenue and data that deliberately subsidize and de-risk lower-maturity ecological and planetary modules through shared infrastructure and cross-domain learning.

ETHICAL-BY-DESIGN & REVERSIBILITY

Non-negotiable guardrails (do-no-harm, reversibility, planetary boundary thresholds) encoded directly into simulation, decision-support, and deployment orchestration layers from day one.

DUAL-USE & MOONSHOT OPTIONAILITY

Immediate clinical and terrestrial value creates sustainable economics while building the exact capabilities required for long-duration space life support and eventual controlled planetary engineering.

OPEN PLATFORM WITH STRATEGIC SAFEGUARDS

Standards and kernels developed openly to accelerate collective progress while protecting biosafety, national security interests, and core IP where strategically necessary.

CUSTOMER SEGMENTS & GO-TO-MARKET — PRECISE TARGETING

BIOTECH & PHARMA

GTM: Enterprise licensing + co-development partnerships for cell/gene and tissue engineering pipelines. Value: reduced development time & cost, higher success probability via digital twin evidence.

Channel: Direct sales + strategic alliances

HEALTHCARE SYSTEMS & PAYERS

GTM: Platform subscription for personalized therapy planning and outcome prediction. Value: lower late-stage costs, better patient outcomes, evidence for value-based contracts.

Channel: Enterprise SaaS + outcome-based contracts

GOVERNMENTS & NGOs (RESTORATION)

GTM: Landscape-scale adaptive management platforms + verification infrastructure for biodiversity credits and carbon markets. Value: standardized, auditable, adaptive control.

Channel: RFPs, cooperative agreements, UN Decade alignment

SPACE AGENCIES & COMMERCIAL SPACE

GTM: Bio-regen + ISRU hybrid modules for Artemis, Mars missions, commercial stations. Value: closed-loop life support, reduced resupply mass, mission resilience.

Channel: Multi-year contracts, joint development

RESEARCH CONSORTIA & UNIVERSITIES

GTM: Academic licensing + consortium participation for simulation kernels and standards development. Value: shared infrastructure, publication acceleration, talent pipeline.

Channel: Consortium membership + data access

IMPACT & CLIMATE FUNDS

GTM: Strategic investment + governance advisory for planetary-boundary aligned portfolios. Value: measurable, verifiable regeneration outcomes at scale.

Channel: Direct LP relationships + co-investment

REVENUE STREAMS & UNIT ECONOMICS — PRECISE MODEL

PLATFORM LICENSING (CLINICAL + ECO)

Annual or milestone-based licensing of core modules and simulation kernels. High gross margin once developed. Early customers: biotech companies and large restoration programs.

Timing: Phase 1 ramp

SAAS — DIGITAL TWIN & ORCHESTRATION

Subscription access to multi-scale digital twin platform, generative design engine, and real-time orchestration. Tiered by scale (cell/tissue vs landscape vs planetary).

Timing: Phase 1-2 growth

DATA MARKETPLACE

Anonymized, consented outcome data from pilots and deployments sold to researchers, insurers, and model trainers. Privacy-preserving by design.

Timing: Phase 2+

GOV & SPACE AGENCY CONTRACTS

Multi-year R&D and deployment contracts for bio-regen modules, analog testbeds, and governance frameworks. Stable, high-value, strategic anchor revenue.

Timing: Phase 1-3

MILESTONE / SUCCESS-BASED PAYMENTS

Performance-linked payments from pilots and large deployments (e.g., restoration success metrics, therapy outcome improvements). Aligns incentives.

Timing: Phase 1+

GOVERNANCE & ETHICS ADVISORY

Retainer and project-based advisory for governments, space agencies, and large NGOs on planetary intervention ethics, reversibility protocols, and multi-stakeholder governance design.

Timing: Phase 2-3

COST STRUCTURE & PHASED BURN RATE — PRECISE FINANCIAL DISCIPLINE

PHASE 1 (2026-2029)

High (70-80% R&D)

- Core platform development (AI, simulation kernels, orchestration)
- Bionanorobot sandbox & safety systems
- First clinical + terrestrial pilots
- Core team build (AI, biomathematics, synthetic bio, ecology, governance)
- Initial compute infrastructure & regulatory engagement

PHASE 2 (2030-2034)

Moderate (50-60% R&D)

- Scaling simulation infrastructure & analog testbeds
- Expanded pilot network (clinical + eco + space analogs)
- International governance operations
- Talent acquisition in key domains
- Data marketplace & standards maintenance

PHASE 3 (2035-2040+)

Lower % R&D, higher deployment

- Global deployment & regional testbed operations
- Large-scale governance & ethics institute
- Continuous platform evolution & security
- Partnership & JV management
- Public transparency & stakeholder engagement

REVENUE FORECAST 2026–2035 — MIN / BASE / MAX (USD MILLIONS)

Assumptions: Phase 1 pilots convert to licensing & grants. TRL Pull accelerates adoption. Base = successful clinical pilots 2028 + first eco contracts 2030. Max = major space anchor contract by 2032. All figures illustrative forward-looking.

SCENARIO	2026	2027	2028	2029	2030	2032	2035
Revenue — Base	0.8	2.5	6.2	14	28	85	240
Revenue — Min	0.4	1.2	3.0	7	14	40	115
Revenue — Max	1.5	5.0	12	25	55	160	480
Gross Margin (Base)	25%	32%	38%	45%	52%	60%	68%

KEY INSIGHTS: Base case reaches meaningful revenue scale by 2029-2030 as Phase 1 pilots convert to recurring licensing. Max scenario assumes early major space or climate fund anchor contract. Gross margins improve steadily as platform scales and fixed simulation costs are amortized.

Note: Illustrative only. Actual trajectory depends on pilot success, regulatory pathways, and partnership execution. Conservative relative to comparable deep-tech platform ramps.

CAPEX BREAKDOWN & INFRASTRUCTURE INVESTMENT ROADMAP

CAPEX CATEGORY	Phase 1 (26-29)	Phase 2 (30-34)	Phase 3 (35-40)	NOTES
AI / Simulation Compute Clusters	\$8-12M	\$15-20M	\$10-15M	Physics-informed NN training, real-time twin inference, generative design workloads. Cloud + on-prem hybrid.
Bionanorobot R&D Lab & Safety Systems	\$4-6M	\$3-5M	\$2-4M	Cleanroom, characterization, containment, kill-switch validation, progressive sandbox infrastructure.
Analog Testbed & Field Deployment	\$2-4M	\$8-12M	\$6-10M	Terrestrial extreme environment sites (Arctic, Atacama, Antarctic) + controlled bioregenerative modules for space analogs.
Governance & Ethics Infrastructure	\$1.5-2.5M	\$3-5M	\$4-6M	Multi-stakeholder platforms, transparency dashboards, international coordination systems, audit & verification tooling.
Data & Standards Infrastructure	\$1-2M	\$2-4M	\$2-3M	Extended FHIR + ecological metadata, synthetic bio ontologies, secure data exchange, marketplace backend.

Total illustrative CAPEX across all phases: ~\$70-110M (cumulative). Front-loaded in Phase 1-2 to build durable platform and testbed assets. Later phases shift toward deployment and operations.

OPEX BREAKDOWN & TALENT / OPERATIONS RAMP — PRECISE SCALING

OPEX CATEGORY	Phase 1	Phase 2	Phase 3	KEY NOTES
Core Technical Talent (AI, Biomathematics, Synthetic Bio, Ecology, Space Systems)	12-18 FTEs	35-50 FTEs	60-80 FTEs	Highest cost and highest leverage. Competitive compensation + mission equity.
Governance, Ethics & Regulatory Team	3-5 FTEs	8-12 FTEs	12-18 FTEs	Critical for sandbox credibility and international acceptance. Includes ethics board secretariat.
Pilot Deployment & Field Operations	4-6 FTEs	12-18 FTEs	20-30 FTEs	Clinical coordinators, eco-restoration managers, analog testbed leads, safety officers.
Compute, Cloud & Data Infrastructure (ongoing)	\$1.5-2.5M/yr	\$4-7M/yr	\$6-10M/yr	Variable with simulation load. Hybrid cloud strategy with reserved capacity for large runs.
Compliance, Legal, IP & Insurance	\$1.2-2M/yr	\$2.5-4M/yr	\$3.5-5M/yr	Regulatory filings, biosafety, IP prosecution, D&O/E&O, clinical trial insurance, international coordination.

Total illustrative annual OPEX at scale (Phase 3): \$45-70M. Majority is talent and specialized operations. Platform leverage improves unit economics over time as fixed costs are spread across growing revenue base.

EBITDA, BREAKEVEN & CASH POSITION — MIN / BASE / MAX SCENARIOS

METRIC (USD M)	2028	2029	2030	2032	2035
EBITDA Base	-12.1	-6.5	+0.5	+40.5	+179.5
EBITDA Min	-11.0	-9.0	-6.0	+12	+65
EBITDA Max	-4.0	+6	+25	+95	+340
Cumulative Cash (Base, post-funding)	-35	-42	-38	+25	+320
Breakeven Timing	—	—	Late 2029 / 2030	Sustained	Strongly positive

Base case reaches operating breakeven in late 2029 / early 2030 as Phase 1 pilots convert to recurring revenue and TRL Pull begins to compound. Max case (major space/climate anchor) reaches sustained profitability significantly earlier with stronger cash position for accelerated moonshot execution.

Cash position assumes successful fundraising in each scenario. Conservative view — actual capital efficiency expected to improve with platform leverage and partnership leverage.

CAPITALIZATION SCENARIOS — PRECISE USE OF FUNDS & MILESTONES

MINIMUM (\$12-15M SEED)

USE OF FUNDS

Core platform MVP (AI orchestration + basic multi-scale twin engine)
1-2 early clinical pilots + 1 terrestrial eco demo
Core team (15-20 FTEs) + key advisors
Initial governance sandbox & ethics framework
18-24 month runway to first revenue

KEY MILESTONES

Platform v0.8 live + first paying pilot
2 peer-reviewed publications from pilots
TRL Pull mechanism validated internally
Seed-to-Series A bridge ready by 2028

VALUATION & ROI

Post: \$45-60M
Series A target: \$25-35M (2030)
Base ROI: 8-12x by 2035

MEDIAN (\$35-45M SERIES A)

USE OF FUNDS

Full Phase 1 execution (platform v1.0 + bionanorobot sandbox)
4-5 validated pilots across clinical + eco
Expanded team (40-50 FTEs) + international presence
Major compute + analog testbed partnerships
Governance board & regulatory engagement

KEY MILESTONES

3-5 peer-reviewed use cases with measurable outcomes
First cross-domain simulation validation published
2+ gov/space agency LOIs or contracts
Ready for Series B or strategic (2030-31)

VALUATION & ROI

Post: \$140-180M
Series B / strategic: \$60-90M
Base ROI: 15-25x by 2035

MAXIMUM (\$70-90M+)

USE OF FUNDS

Aggressive multi-domain build + space track acceleration
Dedicated Mars analog + ISRU bio-hybrid program
International governance leadership role
Large-scale talent + complementary IP acquisition
Global pilot network (3 continents) + ethics institute

KEY MILESTONES

Platform v1.5 with full cross-scale interoperability
Anchor contract with major space agency or climate fund
First regional Earth engineering testbed launched
De-facto standard position by 2032

VALUATION & ROI

Post: \$280-400M+
Strategic / IPO path 2033-34
Base ROI: 25-40x+ by 2035

INVESTOR ROI, EXIT PATHS & LONG-TERM STRATEGIC VALUE — MOONSHOT THESIS

INVESTMENT THESIS & ROI DRIVERS

- Platform moat: Proprietary multi-scale simulation kernels + cross-domain ontology create durable data and IP advantage that compounds with every deployment.
- TRL Pull flywheel: Early clinical revenue and regulatory precedent subsidize higher-upside planetary and space modules — asymmetric risk/reward profile.
- Dual-use optionality: Clinical and terrestrial success de-risks; space and large-scale eco success provide 10x+ tail scenarios.
- Governance premium: First-mover advantage in ethical, multi-stakeholder framework for planetary intervention creates regulatory moat and preferred access to public-sector contracts.
- Comparable returns: Similar deep-tech platforms in synthetic biology, climate AI, and space infrastructure have delivered 15-40x returns to early investors on successful execution paths.

Base case target (Series A entry): 12-20x return by 2035. Max case: 30x+ with major space anchor or climate fund outcome.

EXIT PATHS & STRATEGIC VALUE

Strategic Acquisition (primary path 2032-2035):

- Big Pharma / Biotech seeking digital twin + regeneration platform
- Climate Tech / Impact majors seeking landscape regeneration OS
- Space & Defense primes (bio-regen + ISRU for long-duration missions)
- Big Tech (AI + simulation for life sciences / sustainability)

IPO / Public Markets:

- 2033-2035 window if Phase 2/3 milestones achieved and revenue trajectory >\$100M
- Natural peer set: regenerative medicine platforms, climate tech infrastructure, space economy companies

Government / Strategic Anchor:

- Long-term NASA/ESA or equivalent multi-year contracts provide stable cash flow and valuation floor

Strategic value beyond financial returns: URTE positions investors at the center of the defining technological and ethical challenge of the century — regenerating life on Earth and preparing for multi-planetary civilization.